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Analyze in Depth Health Care Business Process and Key Performance Indicators using Process Mining

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Abstract

Process mining is applied in literature to many case studies in various domains. It represents an explorative procedure, very helpful to the domain experts for analyzing data from multiple perspectives. In this paper, we concentrate on extracting knowledge from event logs related to a health care process that is recorded by a Business Process Management System and mining Key Performance Indicators content associated to process instances data. Thus, this work contributes to the evaluation and the enhancement of the healthcare process using Process mining tool.

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1. Introduction

Business Process Management has many advantages such as enhancing processes to suit evolving business needs and measuring and optimizing processes leading to effective overall company performance and ensuring that the processes still meet the goals for which they were created. In [1], Van Der Aalst et al define Business Process Management (BPM) as a discipline that advocates the use of continuous improvement on processes. This discipline builds upon earlier work on workflow management, operational research, lean manufacturing, six sigma, and business process re-engineering. It also combines knowledge from information technology, management science, and industrial engineering.

BPM involves a series of phases which can be depicted in a continuous life-cycle. In literature, there are different BPM life-cycles proposed by several researchers. The BPM life-cycle [2] defines how to improve processing based on the knowledge of historical executions. In this paper, we focus on BPM lifecycle described by Weske in [3]. In the design and analysis phase, the processes are identified, (re)designed and validated using for example simulation techniques to verify that the process actually exposes the desired behavior. A Business Process Modeling Notation is used in this phase. In the configuration phase, BP model is implemented by configuring an IT-system. The enactment phase incorporates the execution of the business process using the system configured. Using the monitoring component, the administrator can track and visualize the status of each process instances and other information with the help of the IT-systems. In the evaluation phase, the BP is diagnosed and the information is collected to evaluate and improve business process model and its implementation. The last phase involves the continuous monitoring of processes where many techniques can be used such as process mining to analyze the information stored in event data. Process Mining is a collection of techniques which aim to discover, monitor and improve business processes [4].

Business Process Management incorporates Business Process Management Systems (BPMSs) as mechanisms to optimize and automate organizations daily procedures. These BPMSs manage the information related to the instances (business data) related to the execution of the BP model and recover the information concerning the process performance (Process Performance Indicators) [5].

Typically, execution logs generated in the enactment of the BPM phase are used to store information about processes real execution, such as the start or the end of activities. The use of a BPMS facilitates the monitoring of Key Performance Indicators (KPIs). The definition of Key Performance Indicator has drawn the attention of many authors in several domains. For instance, in [6] Villalón et al define an indicator as a metric that evaluates the progress and the performance of the process with respect to some objectives. The authors add later that such metrics can be directly derived from data or can use a formula to combine values. In [7], Kritikos et al state that KPIs usually include a metric and a threshold imposed on it, thus defining the respective minimum performance level to be sustained. In [8], Looy et al note that performance indicator should also have a concretization that makes clear how it is measured, and its value should be compared to a target value.

BPMSs generate and store information about their performance, such as the number of instances of a business process execution, the duration time of each activity, who has executed each activity, the resources involved, the frequency of each activity, and the number of instances [5]. In [9], the authors evaluate each BPMS consistently by calculating the total support level of every BPMS and sorting evaluated systems according to their level of support for all evaluation criteria. The best rate is provided by jBPM followed by Camunda, YAWL, Bonita, Activity, ProcessMaker, and uEngine.

The process considered in this work is the process of healthcare. The proposed business process model is deployed with the use of jBPM software, where 100 instances have been created through the execution of the process. There are many measurements that could be used in the evaluation of healthcare process, but after discussion with experts, we arranged a list of relevant indicators. Our Data comes essentially from BPM database system and estimated KPI files. The ProM process mining software was chosen in this research. This tool has an intuitive interface and allows automatically constructing process visualization.

The remainder of this paper is organized as follows: in section 2, we use process mining to analyze emergency healthcare process. In section 3, we analyze KPI data file. This file contains quantitative and qualitative indicators target values related to each process instance. The last section gives a brief conclusion.

2. Business Process mining

Before we apply process mining we should prepare and clean the log to answer some questions related to the data we are interested in. For example, how many process instances/tasks/resources are in the log? What are the most frequent paths in the process? Are there any loop paths in the process? Thus, in this step, we are based essentially on Bamtasksummary (figure 1) because it contains information about the dates and times on which tasks were executed.

pk [PK] bigint	createddate timestamp without time zone	duration bigint	enddate timestamp without time zone	processinstanceid bigint	startdate timestamp without time zone	status character varying(255)	taskid bigint	taskname character varying(255)
8	2016-05-26 11:44:25.83	134152	2016-05-26 11:46:49.535	155	2016-05-26 11:44:35.383	Completed	485	Registration
9	2016-05-26 11:46:49.524	218437	2016-05-26 11:50:42.52	155	2016-05-26 11:47:04.083	Completed	486	Sorting
10	2016-05-26 11:50:42.515	197293	2016-05-26 12:03:22.647	155	2016-05-26 12:00:05.354	Completed	487	Consultation in box
11	2016-05-26 12:03:22.643	9322161	2016-05-26 15:00:57.104	155	2016-05-26 12:25:34.943	Completed	488	Lying waiting
12	2016-05-26 12:33:44.993	73510	2016-05-26 12:35:09.739	156	2016-05-26 12:33:56.229	Completed	489	Registration
13	2016-05-26 12:35:09.736	269042	2016-05-26 12:41:52.786	156	2016-05-26 12:37:23.744	Completed	490	Sorting
14	2016-05-26 12:41:52.779	1050390	2016-05-26 13:00:35.476	156	2016-05-26 12:43:05.086	Completed	491	Consultation in box
15	2016-05-26 09:37:12.317	79784	2016-05-26 09:38:48.809	157	2016-05-26 09:37:29.025	Completed	492	Registration
16	2016-05-26 09:38:48.806	82116	2016-05-26 10:03:26.327	157	2016-05-26 10:02:04.211	Completed	493	Sorting
17	2016-05-26 10:03:26.322	332223	2016-05-26 10:56:15.236	157	2016-05-26 10:50:43.013	Completed	494	Consultation Delayed emergency

Figure 1. Screenshot of BAMTaskSummary Table

To do that we import to ProM the corresponding event log which contains essentially activities that happen, the process instance identifier, and the start/end date timestamp. Based on the “log visualizer”, we especially focus on the “Summary” tab and we get the following visualization (figure 2).

Event Name	
Event classes defined by Event Name	
All events	
Total number of classes: 6	
Class	Occurrences (absolute)
Registration	100
Sorting	100
Consultation in box	86
Lying waiting	19
Consultation Delayed emergency	12
Shock treatment	3
Start events	
Total number of classes: 1	
Class	Occurrences (absolute)
Registration	100
End events	
Total number of classes: 4	
Class	Occurrences (absolute)
Consultation in box	67
Lying waiting	19
Consultation Delayed emergency	12
Shock treatment	2

Figure 2. Screenshot of process log summary related to the EC process

With ProM, we can also inspect activities occurrences from the event logs. In this view, the user will get overview information about the number of different classes of registered execution scenarios. We can consider the role of this log view like the dash builder module provided by jBPM because both of them provide some performance information about the process. The coloring of activities provided by the software is very important to see for example how frequently each activity occurs or the main paths of the process flow (figure 3).

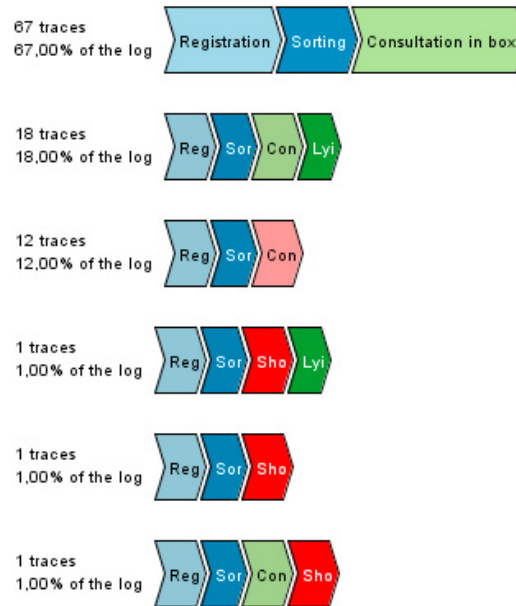


Figure 3. Screenshot of possible paths related to the EC process

Based on this view (figure 3), our analysis is based essentially on the completed process instances only. In this way, the decision maker can see how the process has been actually performed and the most frequent completed paths and also can analyze the distribution of all cases over the different paths through the process. We can see that all cases start with the registration activity. Afterward, all patients enter the sorting room for primary exams. In conclusion, we can say that the main process flows (67% of the log) contain 3 activities which are registration, sorting, and consultation in box. If we have more complicated BP and more BP traces we can see for example which process steps happened multiple times? Which activity takes a lot of time before starting? How many actors are involved in a specific case? In this way, it becomes easy to detect paths or activities that cause a bottleneck. The more data attributes we have in the logs, the more additional statistics we get. These data are very important for the process analysis because they hold relevant context information such as: By which actor the activity was handled, in which department the process was performed, etc.

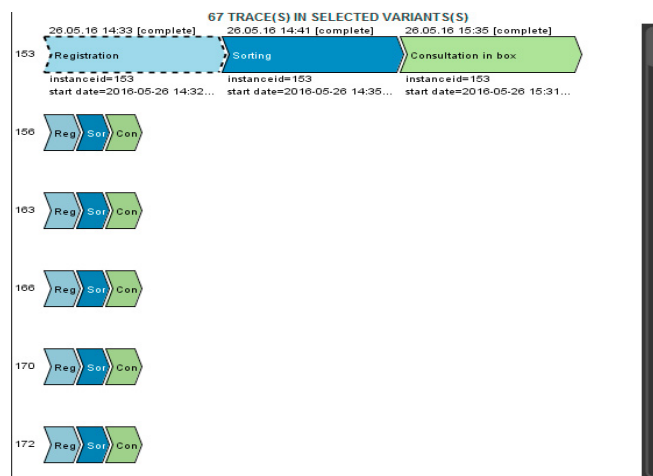


Figure 4. Screenshot of some cases related to the selected flow

We can find out more information when we inspect this log. For example, we can inspect the process instances and browse the set of cases in the log (figure 4). We can also see the events in a selected case to get a detailed view. For example, figure 4 shows information related to process instance id number 153.

Second, using process mining tools, we can construct a more critical vision based on Process discovery in order to suggest a process remodeling or to analyze gaps in the process. The process mining software supports various actions to mine the control flow perspective of process models in order to see how cases are actually being executed in the organization. For example, when we use the “ α -algorithm” plug-in, it allows to visualize a Petri Net graph and then we can convert this graph to a BPMN diagram. In this way, we obtain the process model that summarizes the sequences of activities followed by most/all cases in the log.

Based on all available information, we can set the following points:

- The process takes too much time for patients who have a delayed emergency. Especially, it takes an important waiting time before the delayed emergency starts.
- The process takes too much time for patients who are oriented to the supervision room. Especially, it takes an important time to lying waiting activity to end.
- In some cases, the consultation activity takes too much time to end.

When we review these observations with the process stakeholders, we conclude the following explanation:

- From the process manager view, patients who are in the delayed emergency can wait because they are not urgent.
- For the considerable time occupation in the lying waiting, it is explained by the fact that the health of these patients is not stable.
- For the important time taken in the consultation in box, it is explained by the fact that some patients required complementary exams which are frequently made in other departments of the hospital.

Third, in order to improve business processes and KPIs, we consider the mining of KPIs as an explorative and highly interactive task where the decision makers learn to understand the data in two ways. First by understanding KPI relationships retrieved from the knowledge database and second by using some data retrieved from KPI estimated file and process instance log in order to look at the process from multiple views using ProM software.

3. Key Performance Indicator mining

We proceed by using ProM software. In this step, we analyze KPI target values retained from KPI estimated file (figure 5) and we combine data with the appropriate rows and columns retrieved from process instance log table by using the same process instance identifier. The KPI file “estimated values” contains all evaluated data after comparing the real value with the target value or threshold value set by the decision maker. This file contains two values (“ok” codes the KPI value which is considered by the organization as acceptable and “not ok” corresponds to the fact that the KPI value is not acceptable).

Quali_KPI16	Quali_KPI17	Quali_KPI18	Quali_KPI19	Quanti_KPI1	Quanti_KPI1	Quanti_KPI2	Quanti_KPI3
ok	ok			ok	Not ok	ok	
ok	ok	ok		ok	ok	ok	Not ok
ok	ok			ok	ok	Not ok	
Not ok	Not ok			ok	ok		
ok	ok	ok		ok	ok	ok	ok
ok	ok			ok	ok	ok	ok

Figure 5. Screenshot 2 of some KPI estimated values

After preparing the data set, we import the file into ProM. We can get further analysis by analyzing some relationships between indicators. For example, we analyze the level of satisfaction Quanti_KPI9 and the waiting time before sorting with Quali_KPI13 installation in the sorting room. The aim of this analysis is to check if the waiting time before sorting, respects business rules (i.e. evaluated as OK value), also we check if the qualitative indicators related to this quantitative indicator retained from patient experience are respected (i.e. satisfaction of the

patient). To do that, we focus on log visualization to inspect some statistics about it. Figure 6 shows the corresponding results.

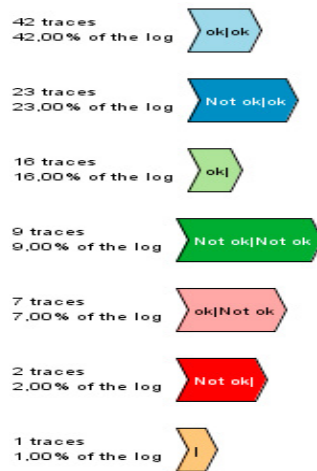


Figure 6. A screenshot of statistics related to Quanti_KPI9 with Quali_KPI13

In this figure, we can observe that 42% of patients are satisfied to very satisfied about their installation in the sorting room and the waiting time before sorting also respects the target value (ok value). The waiting time before sorting doesn't respect the target value for 23% of cases, however, patients are satisfied to very satisfied with their installation in the sorting room. We can also note that 16% of cases have a neutral response about this qualitative aspect and the waiting time before sorting respects the target value. Now if we want to get some statistics in the case where the quantitative and the qualitative indicators are not respected, we can find 9% of the log file. We can inspect this statistic more deeply by browsing the set of cases in the log (figure 7). For instance, we can obtain a detailed view including the corresponding process instances identifiers related to the case where the quantitative and the qualitative indicators are not respected (green color).

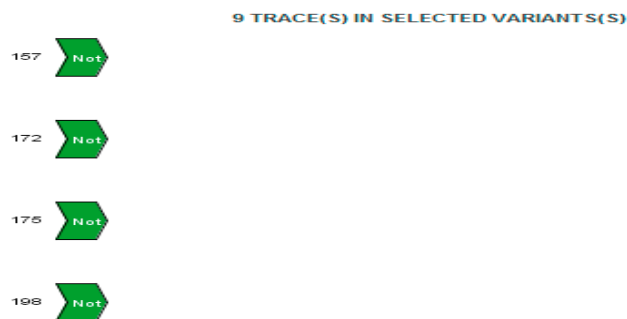


Figure 7. A screenshot of some process instances identifier related to the selected variant

Using ProM and the appropriate file, we can combine as many attributes (indicators) as we want to get some interesting statistics that help decision maker to evaluate the performance of the process. We take for example the combination of Quali_KPI3 paramedical staff availability, Quali_KPI4 medical staff availability and Quanti_KPI9 the waiting time before sorting; we obtain the following traces (figure 8).

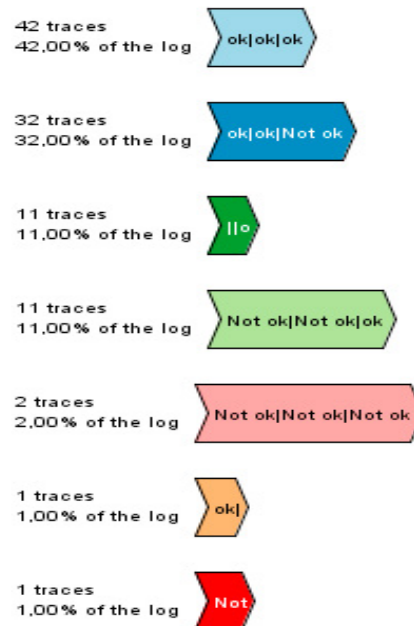


Figure 8. A screenshot of statistics related to Quali_KPI3, Quali_KPI4, and Quanti_KPI9

In figure 8 we focus on four interesting cases. The first one, when all indicators in question respected the desired value (all of them are ok) which takes the major found percentage (42%). The second one, when patients are satisfied to very satisfied about the qualitative aspect related to the availability of medical/paramedical staff with the quantitative aspect related to the waiting time before sorting which is not ok (did not respect the target value). We can find in this category 32% of the log. The third one, when the patient is not satisfied to very dissatisfied (not ok value) about the qualitative aspect with the waiting time before sorting which respects the target value, we found 11% of the log in this category. The fourth one, in the case that this combination didn't respect the desired value which is a very rare case (takes only 2% of the log). Like the first example, we can browse cases to verify if there really exists some patient cases that attend less than 5 min before entering the sorting room (Quanti_KPI9 the waiting time before sorting is ok). However, when they entered to the sorting room, neither paramedical staff nor medical staff is available (Quali_KPI3 paramedical staff availability and Quali_KPI4 medical staff availability are not ok). This case is infrequent, only 11% cases were found, but it is important to see concrete examples particularly for "strange" behavior and to help process stakeholder to find things that are hard to believe until they have drilled down to an individual example case.

To summarize, the majority of patients were satisfied during their experience in the emergency department and the majority of quantitative indicators respect the desired target values based on process instances recorded in the system. However, in rare cases, we observe some deviations. As an example, BP doesn't take too much time but patients already had a bad experience. For example, the delayed emergency duration respect the target value. However, patients are not satisfied about the clarity of information provided by doctors in this activity. Some patients are not satisfied in the lying waiting because the regularity of doctor visit is not adequate.

We can conclude that process mining software offers different views and different performance information for all activities of the process and resources included in the data set. Furthermore, data are usually very important for the process analysis, because they hold relevant context information and orient the decision makers to set additional questions and goals, such as the following:

- Which type of patient category required more attention?
- In which activity the process was performing (respect the qualitative and the quantitative aspect)

We often need more attributes and more cases that are relevant and can be used in the analysis in order to respond to these questions. After discussion with patients and decision makers we can state below some interesting observations:

- The privacy of the patient is not taken into account in the sorting room.
- The necessity of staff to orient the patient according to his/her healthcare status to the appropriate service location
- The necessity to have a system to ensure that data related to each patient (previous analysis, current healthcare status ...) are collected consistently, both within, and across the healthcare departments. Similarly, diagnosis exam results are missing or overlooked and as a consequence appropriate treatment is delayed or repeated.

4. Conclusion

Based on KPI improvement (better selection of KPIs), different possibilities of BP improvement become possible. More precisely, various BP perspectives may be concerned. For example, organizational, behavioral or informational perspectives may be concerned by such improvement. In fact, based on the most important retained KPIs and their corresponding values, BP improvement may concern the assignment of tasks to actors, the order in which tasks have to be fulfilled or input data that will be taken into account for specific tasks.

Therefore, performance indicators can be searched according to that occurrence. Furthermore, decision-makers prefer describing performance measurement indicators that are relevant to their BP. In most cases, decision makers can include managers who don't have to know the details of the indicator. In short, there are still pressing needs for a view that can support performance measurement and BP activities.

Globally, we conclude that the majority of patients were satisfied to very satisfied in the emergency department. However, improvement suggestions are essential for this process, as they enable patients and healthcare providers to effectively identify where the performance of healthcare services requires improvement.

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